DIPLOMA PROJECT

JANTOK Food family meal subscription and coordination system

First stage analysis and design report

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Report status

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The project defines a household account with separate dietary profiles, a visible family subscription quote, meal changes and delivery status. This report consolidates the problem evidence, research, alternatives, requirements, architecture, prototype and evaluation plan. The current prototype is static; operational integrations and measured user outcomes remain later work.

Reading guide

This report supports a first-stage review of feasibility and scope. The first four sections establish the need and market context; sections five to eight explain the chosen design; sections nine to eleven state how the system will be verified, managed and reviewed. Requirements and test identifiers point to the tagged repository snapshot. Statements from interviews and the oral supervisor discussion are identified as team-reported evidence.

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Package navigation

Repository: https://github.com/Balgyn15/jantok-food-diploma

Submission tag: https://github.com/Balgyn15/jantok-food-diploma/tree/prob-period-1

Task board: https://balgyn.youtrack.cloud/projects/JANTOK/issues

Project proposal

Project identity

JANTOK Food is a proposed family meal subscription and coordination system for Astana. The primary users are working adults who plan meals for several household members. Kitchen and operations staff form a second user group. The scope is one kitchen and local delivery; the design does not presume nationwide operation or clinical nutrition advice.

Problem statement

Working households must repeatedly decide what to eat, buy ingredients or place separate orders, prepare food and account for different family preferences and restrictions. After work, this adds time and coordination effort. Ready-meal delivery already exists in Astana, but an integrated household workflow must keep each person's questionnaire, suitable meal alternatives, subscription total and delivery state consistent.

JANTOK addresses this software and coordination problem through one household account with separate member profiles and a shared subscription workflow. The problem is defined clearly; the frequency, severity and preferred solution among Astana households require documented user evidence. The team's recalled informal discussions and their limits are recorded in RELEVANCE_EVIDENCE.md.

Users and stakeholders

- Household owner: organizes meals and manages the subscription for several people.
- Household members: have distinct preferences and restrictions represented in their profiles.

- Kitchen/operations staff: maintain meal information, prepare a daily plan and update fulfillment status.
- Supervisor: reviews the engineering scope, analysis and evaluation design.

The primary user group is working adults and families in Astana. The project does not assume all families can afford or want daily meal subscriptions.

Aim and objectives

Aim: design and evaluate a software system that makes ordering and coordinating ready-made family meals understandable, consistent and traceable.

1. Document household needs and distinguish observations from assumptions.

2. Critically review relevant research and compare existing local services.

3. Define prioritized functional/non-functional requirements and acceptance criteria.

4. Select technologies and define component, data and access boundaries.

5. Design mobile scenarios and the staff interaction needed for fulfillment.

6. Define verification cases and compare usability/coordination outcomes with a simple baseline.

Scope

Included	Outside this submission / core scope
One kitchen and delivery within Astana	Nationwide or multi-kitchen marketplace
Household account, separate member questionnaires and restrictions	Clinical diagnosis, treatment or guaranteed health outcomes
Eligible meal choices and visible reasons for restrictions	Unverified automatic medical recommendations
Itemized 30-day quote and subscription calendar	Production payment processing at the first pre-defense
Meal changes within a stated deadline and order status	A separate courier application
Staff menu/order management and daily kitchen plan	Production provider contracts and commercial launch

Engineering complexity

The system must preserve household ownership, separate member data, consistent eligibility decisions, deterministic quote arithmetic, versioned prices/deadlines and authorized status transitions. Mobile and staff clients share a backend contract and persistent state. Verification covers both business rules and the complete customer-to-staff scenario.

Expected outcomes and success evidence

By 24 October: the analysis/design package, editable diagrams, static prototype, test/evaluation plan, responsibilities and traceable repository/YouTrack records. Subsequent implementation will be evaluated using correct restriction decisions, reproducible quote totals, account separation and successful user task completion against a paper/spreadsheet baseline.

Reduced pressure to cook after work is an intended benefit. Prevention of household arguments is a possible motivation reported by the team, not a measured or guaranteed project outcome. Scope changes and review comments will be recorded in the decision and feedback logs.

Evidence of relevance

Evidence status and method

The project owner reports six real conversations conducted on 15 October 2026. Participant-level summaries were supplied later and published as anonymized paraphrases with the owner's authorization. The owner reports consent from all six participants. Original recordings and consent forms have not been independently examined. An earlier aggregate recollection of four conversations may overlap with these six and is not added to the sample count. The notes describe qualitative observations, not a representative survey of Astana households.

Participant notes

Each entry is a paraphrase of one supplied write-up. The participant codes do not identify people. Frequencies and time estimates are self-reported, not measured. No sentence below is presented as a verbatim quote.

Code	Current meal process and recent difficulty	Preferences, constraints, or reaction	Evidence limit
P01	Usually prepares simple dinners jointly and uses grocery delivery. An unusually late workday led to ordering a prepared meal. Reports that this kind of difficulty is infrequent.	Different food preferences are handled by keeping a shared base and changing one component. Would like more healthy prepared options.	Describes an occasional problem, not regular demand for a subscription.
P02	One household member handles most planning, shopping, and cooking. Reports recurring difficult evenings and roughly two hours a day spent on these tasks.	Different household preferences increase the burden. A previous subscription experience was unsatisfactory because meal quality declined during the week.	Self-reported frequency and time; no observation or price test.
P03	Plans groceries in advance and normally cooks quickly. A temporary household disruption was solved by eating out. Reports little routine difficulty.	Sometimes uses ingredient kits for convenience.	A counterexample to the claim that meal planning is difficult for every target user.
P04	A multi-person household often disagrees about what to eat or who will cook. A missed preparation step led to an improvised meal. Reports frequent time pressure.	Ingredient constraints and price make ready-meal delivery less attractive.	The write-up does not establish which service design would resolve the problem.
P05	Shares cooking duties and uses grocery delivery. An unplanned visit was handled with restaurant delivery. Reports that difficulties are occasional.	Flexible household routines already meet most needs.	A counterexample to broad claims of unmet subscription demand.
P06	Care duties and fatigue can leave purchased groceries unused. Reports repeated difficulty preparing an evening meal.	Multiple food preferences, cost, and packaging waste are concerns about prepared delivery.	The write-up supports a problem statement but does not show willingness to subscribe.

Findings supported by these write-ups

Finding	Participant basis	Requirement or decision link	What the finding does not prove
F-01: Evening planning and preparation can consume substantial time in some households.	P02, P04, P06	Motivates testing the planning and subscription flow in FR-03/FR-04.	A quote or calendar will save time, or that this burden is common across Astana.

Finding	Participant basis	Requirement or decision link	What the finding does not prove
F-02: Different meal preferences and ingredient constraints can complicate one household order.	P01, P02, P04, P06	Motivates separate member profiles and questionnaires in FR-01/FR-02; informs the permitted alternative flow in FR-04.	Any particular restriction has been clinically verified or that the proposed matching logic is safe.
F-03: Convenience competes with concerns about price, quality, variety, and packaging.	Convenience: P01, P03, P05. Concerns: P02, P04, P06.	Supports making the FR-03 price quote visible; menu quality, variety, and packaging remain product questions to test.	An acceptable tariff, retention rate, or willingness to pay.
F-04: Need differs by household and existing routine.	P01, P03, P05 report infrequent difficulty; P02, P04, P06 report recurring difficulty.	Inform NFR-03 pilot recruitment criteria, subject to separate invitation and consent; avoid treating the six write-ups as a representative sample.	Market prevalence or a single solution suitable for all users.

The links above are design motivations. They do not mean participants requested the numbered features. FR-05 order tracking, FR-07 authorization, and NFR-01/02/04/05 remain primarily workflow, safety, and engineering decisions unless later participant or client evidence supports them.

Assumptions and follow-up questions

- Whether a subscription is preferable to the routines or services people already use remains untested.
- The acceptable price, menu variety, packaging tradeoff, and delivery conditions are unknown.
- A reduction in household disagreement is a hypothesis, not a measured outcome.
- The write-ups do not establish how participants would complete the proposed questionnaire-to-quote journey. Use the planned NFR-03 pilot to test that behavior.
- Ask the project owner to check each paraphrase against the original conversation and retain the consent record privately. Do not invent a recording, an interviewer, or a consent method.

Evidence to requirement links

Finding	Participant basis	Design implication	Unresolved question
Repeated evening preparation pressure	P02, P04, P06	Test the calendar and quote in FR-03/04	Does the workflow save time in real use?
Different household preferences and constraints	P01, P02, P04, P06	Separate profiles and restriction checks in FR-01/02	Are ingredient declarations reliable?
Price, quality and packaging concerns	P02, P04, P06	Show total price and retain product questions	What price and variety would be acceptable?
Low routine difficulty for some households	P01, P03, P05	Recruit varied users for NFR-03 pilot	For whom is a subscription preferable?

These links are design motivations, not proof that a participant requested the numbered feature. No claim is made about market prevalence, willingness to pay or a reduction in family disagreements.

Literature review

The targeted review in the repository covers twelve meaningfully used sources: ten scientific publications from 2022–2025 and two official context sources. Metadata and abstracts were checked against PubMed

records on 23 October 2026; seven accessible full-text discussion/limitation sections were also checked. This is a targeted appraisal rather than an exhaustive systematic review. Medical and workplace results are not treated as evidence of local adoption.

Structured source analysis

#	Publication / source	Type and method	Relevant result	Limit and use in JANTOK
1	Saad & Islam, 2025, Computers in Biology and Medicine	Scientific scoping review; 36 of 166 screened studies	Food classification/recommendation approaches vary widely.	Heterogeneous methods and outcomes prevent choosing an algorithm by headline accuracy. Start with auditable constraints and compare later recommendations against a baseline.
2	Yera et al., 2023, IJERPH	Scientific systematic review of food recommenders for diabetes	Reviews algorithms and evaluation practices for a restricted diet.	The full-text review notes difficult ontology reproduction, limited diabetic datasets and inconsistent evaluation. Disease-specific methods cannot validate general family advice; use transparent constraint tests.
3	Vandeputte et al., 2023, Journal of Nutrition	Scientific algorithm study plus online tests (255 adults) and app study (27 adults)	User participation in choosing substitutions improved acceptance in the reported study.	Small app sample and a food-swap setting. Offer an explained, selectable replacement rather than silently changing a meal.
4	De Croon et al., 2025, JMIR mHealth and uHealth	Scientific user-centered workplace deployment; interviews/prototypes/proof of principle	Explanations helped understanding and perceived control; attrition was 77% after seven weeks.	Self-selected users, mixed real/virtual orders and high attrition limit generalization. Workplace catering differs from family delivery. Show exclusion reasons; measure comprehension and retention separately.
5	Towards interactive explanation-based nutrition virtual coaching systems, 2024	Scientific interactive-system experiment	Participants valued feedback and explanations about alternatives.	Explanation and feedback were introduced together, so their separate effects cannot be isolated. Virtual coaching differs from fulfillment. Prototype concise reasons and collect task-level feedback.
6	Sun et al., 2023, Journal of Medical Internet Research	Scientific systematic scoping review of health recommender systems	Only 10 of 51 included systems considered end-user preferences in design/development.	Language/search restrictions may omit studies; the scoping review does not determine outcome utility. Broad health evidence does not establish Astana priorities. Include users in requirement validation.
7	Effects of a personalized nutrition program on cardiometabolic health, 2024, Nature Medicine	Scientific randomized trial, 347 adults, 18-week app intervention	Triglycerides and some secondary outcomes improved; LDL cholesterol did not significantly differ.	Biomarker-heavy intervention and declared commercial ties; JANTOK cannot claim similar health outcomes. Evaluate usable workflows and constraint compliance instead.
8	Effect of a Personalized Diet… vs a Low-fat Diet, 2022, JAMA Network Open	Scientific randomized trial, 199 analyzed	Personalized diet did not yield significantly greater weight loss at six months.	Single-system, English-speaking sample, limited retention and pandemic disruption constrain generalization. Clinical participants differ from JANTOK users; personalization cannot be presumed superior.
9	Shin et al., 2025, American Journal of Preventive Medicine	Scientific randomized online-grocery trial	A combined labeling, feedback, ordering and substitution intervention improved cart nutrition score in that setting.	Cannot isolate the effect of substitutions; shopping is not prepared-meal delivery. Compare JANTOK recommendations with a simple unfiltered menu baseline.

#	Publication / source	Type and method	Relevant result	Limit and use in JANTOK
10	Health-Aware Food Recommendation Based on Knowledge Graph and Multi-Task Learning, 2023	Scientific algorithm/model evaluation	Models preference and health-related factors jointly.	One Allrecipes dataset, four nutrient dimensions and no user study limit evidence. Allergies are suggested as future work, not a verified capability. Keep hard exclusions separate from ranking.
11	WHO, Healthy diet, 2026	Public-health guidance, not a project user study	Adequacy, balance, moderation and diversity are broad principles.	Do not convert population guidance into personal medical prescriptions; use it as background and require professional review of nutrition rules.
12	US FDA, Food Allergies	Official food-safety guidance, not local regulation	Cross-contact and undeclared allergens remain risks even when ingredient lists exist.	A software filter is only one control. Require ingredient provenance, kitchen process review and prominent uncertainty warnings before real use.

Count: 12 meaningfully used sources; 10 scientific publications, all dated 2022–2025. WHO/FDA are context and safety sources, not substitutes for research or applicable Kazakhstan requirements.

Synthesis and design implications

1. Preference ranking and allergy exclusion solve different problems. A deterministic exclusion check should run before any preference score (sources 2, 10, 12). A blocked item must show a reason and never be offered as a selectable alternative.

2. Explanations and user choice have promising evidence, but the studies are context-specific and may have high attrition (sources 3–6). The prototype shows meal details, replacement options and an allergy conflict; usability sessions must test whether people understand them.

3. Personalized nutrition has mixed outcome evidence (sources 7–9). The diploma aim should be reliable family planning and traceable fulfillment, with operational and usability metrics, rather than an unproven health benefit.

4. A source does not validate the local problem. The anonymized P01–P06 notes document six conversations reported by the project owner and add local problem observations, including counterexamples. The original conversations and consent records have not been independently checked; the small group does not establish prevalence or demand. Its findings should be tested with users and used to revise requirements and this review when warranted.

Search and appraisal method

Searches on 22–23 October 2026 used PubMed/publisher/official records with food recommender, personalized nutrition, explanation, allergy, digital intervention and randomized trial. Selection favored recent scientific work with an explicit design implication; WHO/FDA supplement context. Clinical studies were retained as counter-evidence to exaggerated personalization claims, not as a recipe for the service. Studies with unrelated populations are explicitly limited in the table.

The search is targeted and does not claim systematic-review completeness. Full-text discussion/limitation checks strengthen appraisal for seven sources; the three record/abstract-only sources cannot establish detailed implementation or bias claims. Further full-text reading and source changes should be recorded as subsequent review work. WHO/FDA do not replace applicable Kazakhstan requirements or kitchen expertise. Each source is meaningfully used in the table/synthesis rather than counted solely as a link.

Resulting design decisions

Hard dietary exclusions must run before preference ranking. A meal with unknown ingredient data should be held for verification, and a blocked item should show a reason. The mobile flow should let a household member inspect a permitted replacement instead of silently substituting. Evaluation should measure whether people understand these decisions. Personalized nutrition studies have mixed outcomes; JANTOK therefore measures workflow clarity, consistent constraints and fulfillment rather than asserting health improvement.

Existing system comparison

Selection and common criteria

Five ready-meal providers whose official material names Astana were checked on 23–24 October 2026. The eight common criteria are city availability, ready meals, separate member profiles, restriction handling, plan duration, meal changes, price visibility and order/delivery status. “Not established” means the inspected public page did not demonstrate the feature; it is not evidence that the service lacks it. No purchase or private account test was performed.

Provider	Plan and price	Dietary handling	Changes	Household and status evidence
FIT MEAL [M01]	7-day example; tariff visible	Adjustment for allergies/intolerance in FAQ	Pause through staff	Multi-member profiles and customer status interface not established
Hemingway Healthy Club [M02]	1/7/10/14/21/30 days; totals shown	Tailoring advertised	Not established	Multi-member profiles and customer status interface not established
Healthy Nation [M03]	1/3/5/7/10/14/21/28 days; example amount	Not established	Visible replacement controls	Multi-member profiles and customer status interface not established
EatFit [M04]	7/14/21/28 days; daily and total price	Allergy/intolerance handling advertised	Meal replacement advertised	Two rations to one address; separate member questionnaires not established
SKINNY zucchini [M05]	Trial/day and 20/28-day offers; daily price	Nutritionist adaptation described	Day transfers and allergy replacements advertised	Multi-member profiles and customer status interface not established

Interpretation and decision

Astana already has direct competitors offering prepared meals, subscription periods, visible pricing and some dietary tailoring. The proposed distinction is the combined software workflow: one household account with separate questionnaires, explicit restriction decisions, coordinated meal choice, an itemized family quote and fulfillment information. Public material did not establish this entire combination for the five providers, but that does not prove exclusivity. Future task-based comparison must test whether the combination solves a meaningful problem and whether households prefer it to existing routines.

Official evidence URLs [M01]–[M05] are listed in References. Advertising claims are described as claims; service performance and account behavior were not independently tested. Displayed tariffs were not ranked because meal sizes, durations and offers differ.

Technology comparison and decisions

Decision method

The team compared each selected component with two practical alternatives against the two-person team’s scope, shared business rules, testability and client needs. These are design decisions for the first-stage baseline, not claims of deployed software or measured platform superiority. Exact package versions and lockfiles belong to the implementation stage.

1. Mobile application

Alternative	Fit for JANTOK	Tradeoff
Flutter / Dart	One component for Android and iOS, consistent questionnaire/calendar UI, unit/widget/integration tests	Team maintains Dart and validates platform plugins on each target

Alternative	Fit for JANTOK	Tradeoff
React Native / TypeScript	React-based mobile UI; language overlap with web/backend	Native module behavior and platform-specific UI still require testing
Native Kotlin/Android and SwiftUI/iOS	Direct platform integration and native UI conventions	Two separate UI implementations increase work for a two-person team

2. Backend platform

Alternative	Fit for JANTOK	Tradeoff
NestJS / TypeScript	Modules, validation/guards and service tests support separate household, quote and staff responsibilities	Runtime validation and authorization still need explicit implementation
Spring Boot / Java	Layered services, mature transaction/test tooling	Adds a separate language and framework setup
FastAPI / Python	Typed request models and API documentation; convenient for data experiments	Python implementation and client contracts require separate coordination

3. Persistent data

Alternative	Fit for JANTOK	Tradeoff
PostgreSQL, constraints	Linked households, members, quotes, orders and immutable snapshots; atomic updates	Requires schema/migration design
MySQL InnoDB	Also provides a credible relational transaction model	Team must coordinate its SQL/migration/deployment conventions
MongoDB	Flexible questionnaire documents with schema validation options	Cross-entity invariants and historical snapshots need deliberate modeling

4. Client/server interaction

Alternative	Fit for JANTOK	Tradeoff
HTTP JSON / OpenAPI	Explicit mobile/staff requests, errors and shared API contract	Endpoint/version discipline required
GraphQL	Clients select fields for complex variable views	Query-level authorization, caching and schema tooling add work
gRPC	Generated typed contracts, especially between services	Browser gateway/toolchain overhead for a small client/API boundary

5. Staff web interface

Alternative	Fit for JANTOK	Tradeoff
Next.js / React / TypeScript	Routing and React components for daily plan, meal and order views	Server/client boundaries and authorization must remain clear
Nuxt / Vue	Structured Vue application and routing	Requires Vue conventions in addition to the selected team stack
Angular	Integrated application structure and forms	More framework-specific concepts for a small staff interface

Supporting choices and change rule

Redis/BullMQ is selected for background processing only when required by notification/provider jobs. Core persistent state stays in PostgreSQL. External payment, SMS and delivery adapters remain boundaries until contracts are available. Exact package versions and lockfiles will accompany runnable component integration. Configuration planning lists manifest locations. Changes to the baseline require a recorded decision, reason and impact on requirements/tests; supervisor approval is recorded separately.

Selected stack and recorded consequences

Boundary	Selected option	Why it fits	Verification focus
Mobile	Flutter/Dart	One shared household UI for Android and iOS	Android first; iOS build and device check separately
Backend	NestJS/TypeScript with Prisma	One API owns restriction, quote, deadline and role rules	Rule and authorization tests
Storage	PostgreSQL	Linked ownership and transactional snapshots	Constraints, migration and idempotency tests
Client contract	HTTP JSON with OpenAPI	Explicit mobile/staff requests and errors	Contract and negative-case checks
Staff web	Next.js/React/TypeScript	Focused kitchen and order interface	Role isolation and status-update tests

Redis/BullMQ may support internal background jobs when notification or provider work is introduced. Payment, SMS and delivery remain adapter boundaries until real contracts and operational rules exist. Changes to the baseline require a dated decision, affected requirement IDs and a verification update.

Requirements specification

The analysis baseline defines eight functional requirements and five non-functional requirements. P0 denotes a mandatory core workflow; P1 denotes supporting capabilities in the planned diploma scope. Acceptance criteria describe observable future behavior. A denied action must leave persistent state unchanged. The rows below do not claim that the functionality is implemented.

Functional requirements

ID	Priority	Expected behavior	Acceptance criterion
FR-01	P0	A household owner creates and manages separate member profiles.	Two members with distinct data are saved and recovered after a new session; editing one does not alter the other.
FR-02	P0	Each member has a separate questionnaire and dietary restrictions; selectable meals respect those restrictions.	For the synthetic fixture, P01 cannot select the milk-containing meal, P02 can, and an item with unknown ingredient data is held for verification. The UI gives the relevant reason.
FR-03	P0	The user receives an itemized 30-day subscription quote before confirming.	Show member/meal quantities, unit prices, currency KZT, period, total, price-rule version and validity. The fixture total is 216000 KZT; changing quantity recalculates it deterministically.
FR-04	P0	A user views the calendar and replaces a meal with a permitted alternative.	An eligible replacement before the configured deadline is saved and visible after reload. An ineligible or late replacement is rejected without changing the plan.

ID	Priority	Expected behavior	Acceptance criterion
FR-05	P0	The household sees current order/delivery status.	An authorized staff status update appears for the correct household and remains after reload; another household cannot read the order.
FR-06	P1	Authorized staff obtain a daily kitchen plan.	Filtering the fixture by date returns the correct order/member/meal quantities and restriction notes. Customer access is denied.
FR-07	P0	Account authentication and role separation protect household and staff actions.	A signed-out caller cannot modify a household; customer A cannot access customer B's member/order; only staff can change fulfillment status.
FR-08	P1	Authorized staff maintain meal ingredients, declared allergens and menu availability.	A changed meal record has a version; future eligibility uses the updated declaration; an existing confirmed plan retains its recorded snapshot.

Non-functional requirements

ID	Priority	Expected quality	Acceptance criterion / measurement
NFR-01	P0	Protect account and dietary information.	Server-side object/role checks cover every private API action; denied accesses return a documented 401/403/404 policy with no private data. Logs/fixtures contain no real participant data.
NFR-02	P0	Preserve price, time and order consistency.	Business deadlines use Asia/Almaty; stored payable totals use integer KZT tenge; repeat confirmation with the same idempotency key creates one logical order. Exact deadline boundary rejects a late change.
NFR-03	P1	Make the core mobile workflow understandable.	At least 4 of 5 pilot users complete the two-member questionnaire-to-quote scenario without hints within 5 minutes; record every failure and device. This is a pilot target.
NFR-04	P1	Support local language use.	The core flow is available in Russian and Kazakh with equivalent field meaning, validation and KZT/date formatting; verify both using the same scripted data.
NFR-05	P1	Respond adequately under a defined small-service load.	For a future staging build, p95 read/quote API response time is at most 1 second with 20 concurrent simulated users for 5 minutes, a fixed 1000-meal/100-household synthetic dataset, server-side timing and recorded hardware. Exclude live-provider calls and report errors separately.

Business rules and dependencies

One kitchen and Astana are the operating boundary. Hard dietary exclusions precede preference ranking. Missing ingredient information cannot be treated as a verified safe match. Quotes record the rule/version and their inputs; confirmed prices do not silently change. The final commercial tariff, change deadline and external provider contracts require stakeholder decisions. Synthetic values in the test fixture are test inputs, not advertised service terms.

Traceability and acceptance

Requirement	Evidence or rationale	Design boundary	Planned cases
FR-01/02	Interview findings F-02; L02/L03/L04/L10/L12	Member/questionnaire, menu eligibility	TC-01/02/03
FR-03/04	F-01/F-03; local plan and change comparison	Quote snapshots and calendar	TC-04/05/06/07
FR-05/06/08	Household/staff fulfillment workflow	Status events, kitchen plan, versioned menu	TC-08/09/10
FR-07; NFR-01/02	Ownership and consistency decision	API guards, transactions, idempotency	TC-11/12/13
NFR-03/04/05	Project evaluation targets	Mobile localization and staging API	TC-14/15/16

The 216000 KZT quote and DEMO-1 rule in the fixture are synthetic test inputs, not a commercial tariff. Time calculations use Asia/Almaty. Ingredient provenance, change deadlines and provider contracts require stakeholder decisions before real service use.

Architecture and diagrams

Architectural overview

The planned system is a modular monolith with a mobile client, staff web client, one authoritative API and relational persistence. The API controls household ownership, restrictions, quote arithmetic, deadlines and order transitions. Provider adapters mark external dependencies. The diagrams are source-controlled with editable Mermaid files and SVG exports; they describe the target design, not an operating deployment.

System context diagram

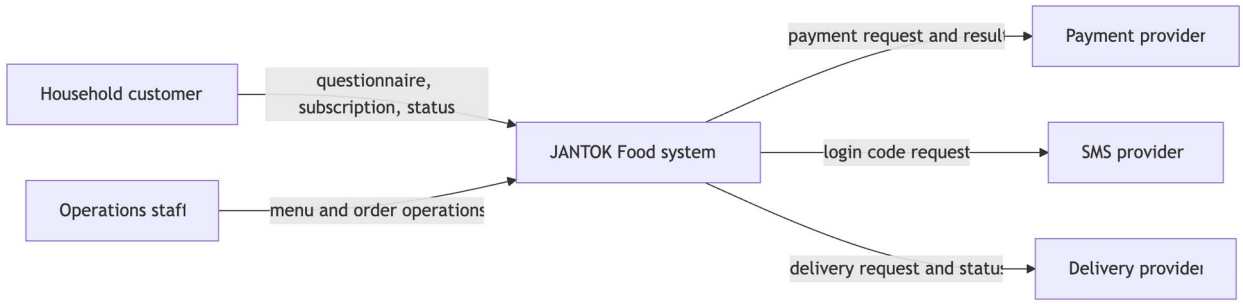


Figure 1. Household and staff users, system boundary and proposed external providers.

The household customer enters questionnaires and subscription choices and reads status. Operations staff manage menus and orders. Payment, SMS and delivery providers are external. Their connections are planned integration boundaries.

Use case diagram

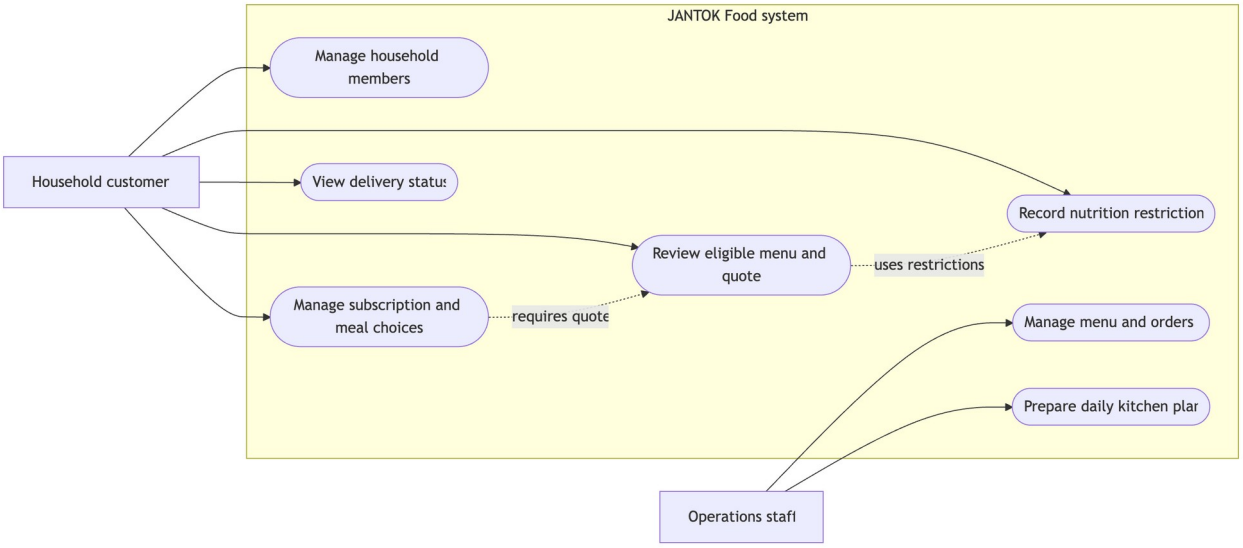


Figure 2. Household and staff use cases with prerequisite restriction checks.

The household owner manages members, restrictions, menu choices, subscription and delivery status. Staff maintain meals, produce a kitchen plan and update fulfillment. Authentication and role authorization are prerequisites for private actions.

Component diagram

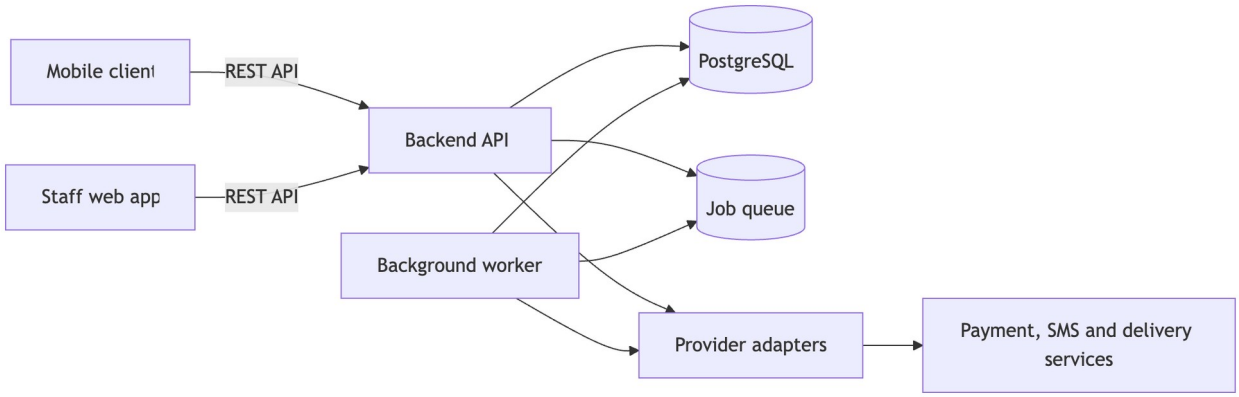


Figure 3. Mobile and staff clients, API, storage, worker and provider adapters.

Flutter and Next.js call the NestJS API over HTTP JSON. PostgreSQL stores authoritative state through Prisma. A worker and queue support retryable background work when introduced; providers are isolated through adapters. This is not a collection of independently deployed microservices.

Responsibilities and data

Boundary	Responsibility	Main data / requirements
Mobile → API	Authenticate owner; manage separate profiles/questionnaires; show eligible meals, quote/calendar and status	Household, Member, Questionnaire, Restriction; FR-01–05/07
Staff frontend → API	Maintain menu data, update authorized status and request daily kitchen plan	Meal/Ingredient/Allergen, MenuVersion, Order; FR-05/06/08
API → PostgreSQL	Preserve ownership and linked state; version inputs for reproducible decisions	QuoteSnapshot, Subscription, MealSelection, StatusEvent; NFR-01/02
Worker → queue/store/adapters	Process a job once logically, retain retries/source, never bypass API-domain rules	Job/Event and provider reference; NFR-02

Boundary	Responsibility	Main data / requirements
Provider boundary	Translate third-party events into checked internal transitions	Payment/delivery reference; contracts pending

Household → Member is one-to-many; a member has versioned answers/restrictions. A quote records quantities, integer KZT prices, period and rule version. A confirmed subscription references its quote and selected menu snapshot. Orders belong to the household and retain authorized status events. Staff access uses a role boundary rather than household ownership.

Main scenario

1. An authenticated owner saves two members and their different restrictions.
2. The API checks ingredient declarations and returns eligible options with reasons for exclusions.
3. The owner selects a 30-day plan and inspects the itemized quote before confirming.
4. Confirmation preserves the quote/menu snapshots; a repeated request is idempotent.
5. A permitted replacement is accepted only before the business deadline.
6. Staff receive the daily plan and update fulfillment; the correct household sees the status.

Requirements, UX, decisions and test cases use the same component responsibilities. Source placement is documented in src. Pricing, nutrition, provider contracts and operational rules require stakeholder confirmation before real service use.

Consistency and security boundaries

Ownership and role checks must happen server-side on every private action. A denied request must return no other household's data and must not mutate state. Quote confirmation must be idempotent, and the business deadline must be evaluated in Asia/Almaty time. These are specified boundaries for future implementation and tests, not behavior demonstrated by the static prototype.

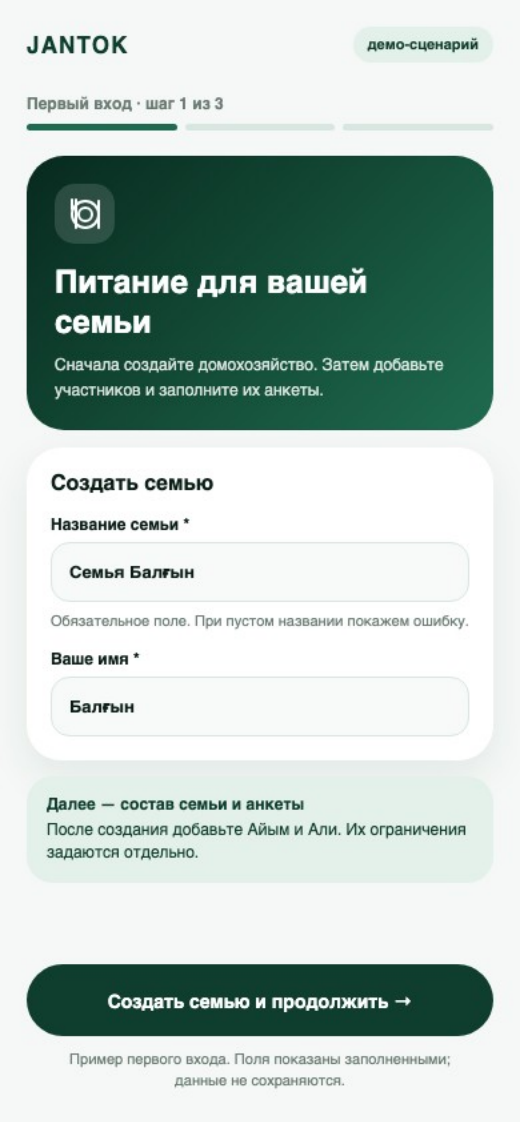
UX and prototype

Prototype purpose and coverage

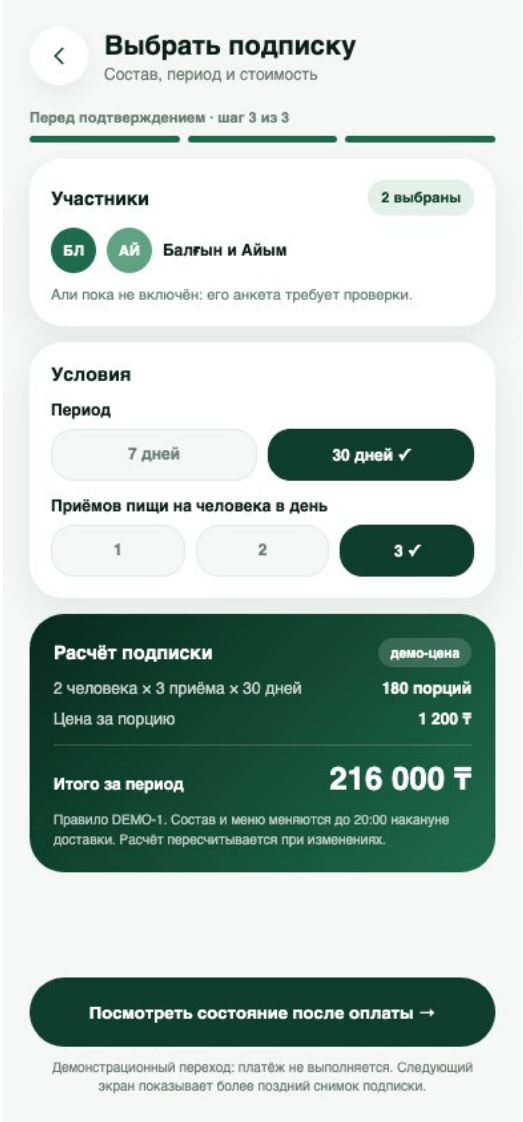
Eight static mobile screens at 390 × 844 pixels cover first use, home, family, questionnaire, menu, subscription choice, subscription summary and delivery. Editable HTML and matching PNG previews are in the repository. The first six screens were supplied from the team's design archive; two additional screens make first use and plan selection visible. Names, prices and order states in the screens are demonstration data. Static controls do not prove navigation, validation, persistence or usability.

Screen	User task	Requirement	What remains to test
00 Onboarding	Create household	FR-01	Empty-name and signed-out errors
01 Home	Review next meal and delivery	FR-04/05	Live status binding
02 Family	Inspect separate members	FR-01	Profile persistence
03 Questionnaire	Set preferences and allergy	FR-02	Saved answers and invalid input
04 Menu	See conflict and alternative	FR-02/04	Safe replacement and deadline
04a Plan choice	Choose period and inspect quote	FR-03	Recalculation and confirmation guard
05 Subscription	Review active plan	FR-03/04	Snapshot consistency
06 Delivery	Read status timeline	FR-05	Authorized live updates

Selected preview screens



First use and household creation



Subscription choice and itemized quote

The plan-selection preview uses a two-member synthetic fixture: 2 members × 3 meals per member per day × 30 days = 180 portions; at 1200 KZT per portion, the total is 216000 KZT. The later subscription screen shows a separate three-member demonstration state and must not be read as the exact result of that quote.

Storyboard and staff interaction

A participant creates a household, adds two members, sets a restriction for one, examines the blocked meal and a permitted alternative, selects a period, reads the itemized quote and locates delivery status. The study records errors and time without coaching. A staff member chooses a kitchen date, reviews meal quantities and restriction notes, and updates authorized fulfillment status; only the owning household sees the resulting status. The staff view remains a planned interface, not a completed dashboard.

The static screens show the intended happy path and some caution states. Actual payment, error handling, data persistence and moderated usability testing are not present in this stage.

Verification and evaluation plan

This plan turns the project aim into measurable acceptance and comparison conditions. No application evaluation result is claimed. The test materials contain planned cases TC-01–TC-16 and a synthetic fixture; executable tests follow runnable features.

Metrics, conditions and baseline

Objective	Metric / target	Measurement conditions	Baseline
Understand household ordering	At least 4/5 pilot users complete the two-member questionnaire-to-quote task unaided within 5 minutes	Five consenting target representatives; same synthetic task and Android device/viewport; count all failures/hints	Paper/spreadsheet household form with the same fields
Understand subscription price	At least 4/5 correctly identify period, quantities, currency and total	Same fixture/catalog; ask before revealing the answer; report each response	Static itemized price sheet with identical inputs
Enforce dietary constraints	100% agreement with independently specified fixture decisions; zero prohibited selectable options in that fixture	Fixed known/incomplete ingredient cases, member separation and invalid input; report dataset size and unknowns	Unfiltered menu with no member-specific exclusions
Coordinate meal/status tasks	Raw completion time, steps and errors for meal change and status lookup; compare each condition	Same order/date/deadline; counterbalance baseline/JANTOK order where possible	Menu spreadsheet plus scripted message/inquiry status scenario
Preserve data and order consistency	All mandatory account/role, quote-snapshot, deadline and idempotency cases pass	Named build, controlled clock, reset synthetic database and logged request/result IDs	Independent expected values and state-transition specification
Respond under small-service load	p95 read/quote server time <=1 second, errors reported separately	20 concurrent simulated users for 5 minutes; 1000 meals/100 households; record hardware/build; exclude provider latency	Same endpoint/build with one user; report load-related change

Targets are project acceptance thresholds. The five-person pilot identifies usability risks and does not establish population-level demand, health benefit or reduced family conflict.

Verification stages

1. Analysis/design now: check requirement coverage, diagram/component consistency, mockup tasks, fixture arithmetic, source links and expected outputs. Keep actual user evidence separate from assumed benefits.

2. During implementation: unit tests for restrictions/quotes/deadlines; API/database integration tests for ownership, snapshots and idempotency; Flutter widget/integration tests; staff-role checks.

3. Integrated evaluation: complete household-to-staff workflow, controlled staging load and moderated usability comparison with the stated baseline.

4. Supervisor review: record comments, resulting corrections and any revised target before evaluating the final build.

Supporting materials and recording

test-cases.csv lists TC-01–TC-16 and links them to requirement IDs. household-scenario.json fixes the synthetic menu, members, price arithmetic and deadline boundary. The tests folder currently contains plans/data; executable tests are added with runnable features.

For each run record case ID, build/tag, date, environment, procedure, expected/actual result, pass/fail, evidence and issue for any defect. For usability retain participant code, task order, device, time, hints, errors and completion. Freeze the fixture and expected values before running; do not change expected outputs to match faulty behavior.

Participation is voluntary and consent precedes observation. Public materials contain anonymous summaries and synthetic data. Keep identifiable notes/recordings outside Git. Software ingredient filtering is not a guarantee against kitchen cross-contact or undeclared allergens; verify those operational dependencies before real use (FDA context).

Test case inventory

Cases	Coverage	Evidence to record
TC-01–03	Member ownership, separate questionnaires and restriction decisions	Stored profiles, allowed/blocked meals and reasons
TC-04–07	Quote arithmetic, snapshots, meal change and deadline	KZT totals, rule version, before/after state
TC-08–10	Order status, kitchen plan and menu version	Authorized status event, date filter, retained snapshot
TC-11–13	Role/object authorization and idempotency	Denied responses, no state change, one logical order
TC-14–16	Usability, localization and small-service load	Participant task record, language checks, p95/error metrics

Detailed planned test cases

Case	Requirement	Procedure	Expected result
TC-01	FR-01	Save two members; update P01; reload the household	Two distinct members persist; P02 unchanged
TC-02	FR-02	Evaluate SYN-M01 for SYN-P01 and SYN-P02	P01 blocked with milk reason; P02 eligible
TC-03	FR-02	Evaluate incomplete ingredient declaration SYN-M03	Hold for verification for both members; not selectable
TC-04	FR-03	Quote 2 members x 3 meals x 30 days x 1200 KZT	216000 KZT; quantities/period/currency/rule version shown
TC-05	FR-03;NFR-02	Save quote then change current price-rule configuration	Existing quote retains original inputs/version/total; new quote uses new rule
TC-06	FR-04	Before deadline choose eligible M02; reload; try blocked M01 for P01	Allowed selection persists; blocked selection leaves prior state unchanged
TC-07	FR-04;NFR-02	Use eligible change at deadline-1s; exact deadline; deadline+1s	Allow first only; reject exact and after; use Asia/Almaty
TC-08	FR-05	Staff updates HHA order status; HHA and HHB request it	HHA sees persistent new status; HHB receives no order data
TC-09	FR-06	Staff filters daily plan for fixture date; customer repeats request	One order; two members; six meals; customer denied
TC-10	FR-08	Staff versions meal declaration; compare future selection and confirmed snapshot	Future eligibility uses new version; existing confirmed snapshot unchanged
TC-11	FR-07;NFR-01	Request private reads/writes as signed-out and another household	Documented denial; no private payload or mutation
TC-12	FR-07;NFR-01	Customer requests staff status update and meal maintenance	Denied; order and meal records unchanged
TC-13	NFR-02	Send same confirmation twice using identical idempotency key	One logical order; repeat returns same result; no duplicate charge
TC-14	NFR-03	Five users perform two-member questionnaire-to-quote task	Target at least 4/5 unaided within 5 minutes; retain all raw outcomes
TC-15	NFR-04	Repeat same core task and validation in Russian and Kazakh	Equivalent meaning/state; correct KZT and date display
TC-16	NFR-05	20 concurrent users; 5 minutes; 1000 meals; 100 households; fixed staging hardware	Target p95 server read/quote time <=1000ms; report errors and hardware; exclude provider calls

All sixteen cases are planned and have not been executed. Their expected results are fixed against the synthetic fixture before a runnable build is evaluated.

The five-person pilot target is a formative usability gate, not a statistical estimate of market demand. A paper/spreadsheet task is the comparison baseline for household coordination; an unfiltered menu is the comparison baseline for restriction decisions. Test reports must give the build, date, device, dataset,

expected result, actual result, error and issue for each case. No pass rate should be stated until tests are run.

Project management and risks

Responsibilities and review

The YouTrack responsibility map assigns research, competitor analysis and UX to Ayana, and the proposal, engineering choices, requirements, architecture, verification and repository planning to Balgyn. Ayana coordinates the supervisor feedback record. The project owner reports that both students personally reviewed one another's materials and that the supervisor/committee can access YouTrack. Individual review and access records were not independently inspected for this report; the assignment map does not itself establish who authored every line of repository content.

Lead	YouTrack tasks	First-stage deliverables
Markhabat Ayana Markhabatkyzy	JANTOK-2, 3, 4, 8; coordinates 11	Interview evidence, critical literature review, five-provider comparison, eight-screen UX, feedback log
Yermakhanbet Balgyn Yerlanqyzy	JANTOK-1, 5, 6, 7, 9, 10	Proposal, technology decisions, FR/NFR specification, three diagrams, evaluation plan, project workflow and risks

Milestones

Stage	Deliverable	Exit condition
First-stage baseline	Analysis/design package at tag prob-period-1	Eleven sections, traceability, diagrams, eight screens, test plan and risks
Foundation and core flow	Identity/household schema, restrictions, menu and quote	Reviewed code and passing core rule/authorization tests
Fulfillment and evaluation	Kitchen/status path, integrations, usability and load assessment	Recorded measured results against the stated baselines

Risk register

Risk	Impact	Response
Interview/consent records are incomplete in public evidence	Interpretation and ethics	Retain private consent record; check paraphrases; avoid prevalence claims
Competing services offer similar plans	Product value	Test the combined family workflow against existing routines
Ingredient declarations or kitchen processes are incomplete	Safety	Block unknown ingredients; obtain operational review before live use
Commercial tariffs and provider contracts are unavailable	Delivery and payment	Use synthetic prices and adapters; defer real integrations
Static prototype omits error and persistence behavior	Usability	Test and implement validation, recalculation and saved state
General oral feedback has no detailed action items	Review traceability	Record date, scope and no-change disposition; capture later specific comments

Repository and submission record

The analysis materials are organized under docs, diagrams, prototype, tests and component-specific src folders. The prob-period-1 tag identifies the first-stage evidence snapshot at commit c59de0d5446b10becdda897a4c2396041a57ed00. Historical pull-request comments and commit dates

should be read as the repository actually records them; task assignments and later review statements should not be projected backward onto earlier merges. The public repository provides the source files and editable diagram/prototype assets.

Supervisor feedback and submission readiness

Supervisor review status

The project owner reported that on 24 October 2026 supervisor Omirzak Islam Amanzholuly orally reviewed the proposal, requirements, architecture and evaluation plan and said that everything was excellent, with no requested revisions. This is the owner's report of an oral conversation; the exact commit reviewed and an independent written record were not supplied. No substantive change is attributed to the general positive remark. Later specific comments, if any, should be logged with their date, affected version and disposition.

First-stage readiness matrix

Area	Prepared evidence	Remaining real-world action
Proposal and relevance	Scope, users, six anonymized notes and limitations	Preserve private consent evidence and verify participant paraphrases
Literature and market	Twelve sources and five-provider, eight-criterion comparison	Recheck changing service claims before final thesis
Technology and requirements	Five stack decisions, eight FR and five NFR	Confirm operational deadlines and commercial inputs
Architecture and UX	Three source/export diagram pairs and eight static screens	Implement and test persistence, errors and staff workflow
Evaluation	Sixteen planned cases, fixture, pilot/load targets	Run on a named build and report measured outcomes
Management and review	Role map, risks, tag and owner-reported oral feedback	Retain personal review/access records where available

First stage submission statement

The tagged repository documents the analysis and design of a household meal subscription workflow for Astana. It includes user-reported problem observations, critical research, evidenced competitor claims, technical and functional decisions, editable system diagrams, a static mobile prototype and an evaluation plan. It does not establish that a full application or provider integrations are operating, that usability targets have been met, or that the service improves health or household relationships. Those claims require later implementation and measured evidence.

References

Research and context sources

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L12. US Food and Drug Administration, Food Allergies. https://www.fda.gov/food/nutrition-food-labeling-and-critical-foods/food-allergies

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M01. FIT MEAL. https://www.fitmeal.kz/

M02. Hemingway Healthy Club. https://hhc.kz/

M03. Healthy Nation. https://healthy-nation.kz/

M04. EatFit. https://eatandfit.kz/

M05. SKINNY zucchini. https://skinny-zucchini.kz/

Platform and security documentation

Flutter. https://docs.flutter.dev/

NestJS. https://docs.nestjs.com/

PostgreSQL transactions. https://www.postgresql.org/docs/current/tutorial-transactions.html

Prisma. https://www.prisma.io/docs

OpenAPI. https://spec.openapis.org/oas/latest.html

Next.js. https://nextjs.org/docs

OWASP object authorization. https://owasp.org/API-Security/editions/2023/en/0x11-t10/